6

INDEX PROCESS WEAKLY INTERACTIVE

- The swap between components i and $i+\epsilon$ of $\mathbf{x} = (x^0, \ldots, x^N)$ occurs with probability

$$\alpha_{i,i+\epsilon}(\mathbf{x}) = 1 \wedge \exp((\beta_{i+\epsilon} - \beta_i)(V(x^{i+\epsilon}) - V(x^i))$$

- By the ergodic theorem we have the acceptance probability at stationarity equals

$$s_{i,i+\epsilon} := \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \alpha_{i,j}(\mathbf{X}_t) = \mathbb{E}[\alpha_{i,j}(\mathbf{X})], \qquad \mathbf{X} \sim \pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$$

- We want to work with the stationary acceptance probability which are independent of $\mathbf{X}_t$

- Suppose π_β is linear path with energy $V = \log \eta - \log \gamma$

$$\pi_\beta(x) = \frac{\exp(-\beta V(x))\pi_0(x)}{Z(\beta)}$$

- Machines only interact weakly by comparing against the energy stored

INDEX PROCESS WEAKLY INTERACTIVE

- Suppose π_β is linear path with energy $V = \log \eta - \log \gamma$

$$\pi_\beta(x) = \frac{\exp(-\beta V(x))\pi_0(x)}{Z(\beta)}$$

  - The swap between components i and $i + \epsilon$ of $\mathbf{x} = (x^0, \ldots, x^N)$ occurs with probability

$$\alpha_{i,i+\epsilon}(\mathbf{x}) = 1 \wedge \exp((\beta_{i+\epsilon} - \beta_i)(V(x^{i+\epsilon}) - V(x^i))$$

  - Machines only interact weakly by comparing against the energy stored

- By the ergodic theorem we have the acceptance probability at stationarity equals

$$s_{i,i+\epsilon} := \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \alpha_{i,j}(\mathbf{X}_t) = \mathbb{E}[\alpha_{i,j}(\mathbf{X})], \qquad \mathbf{X} \sim \pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$$

- We want to work with the stationary acceptance probability which are independent of $\mathbf{X}_t$

MODEL ASSUMPTIONS